Receptive Field Block Net for Accurate and Fast Object Detection

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马栋梁
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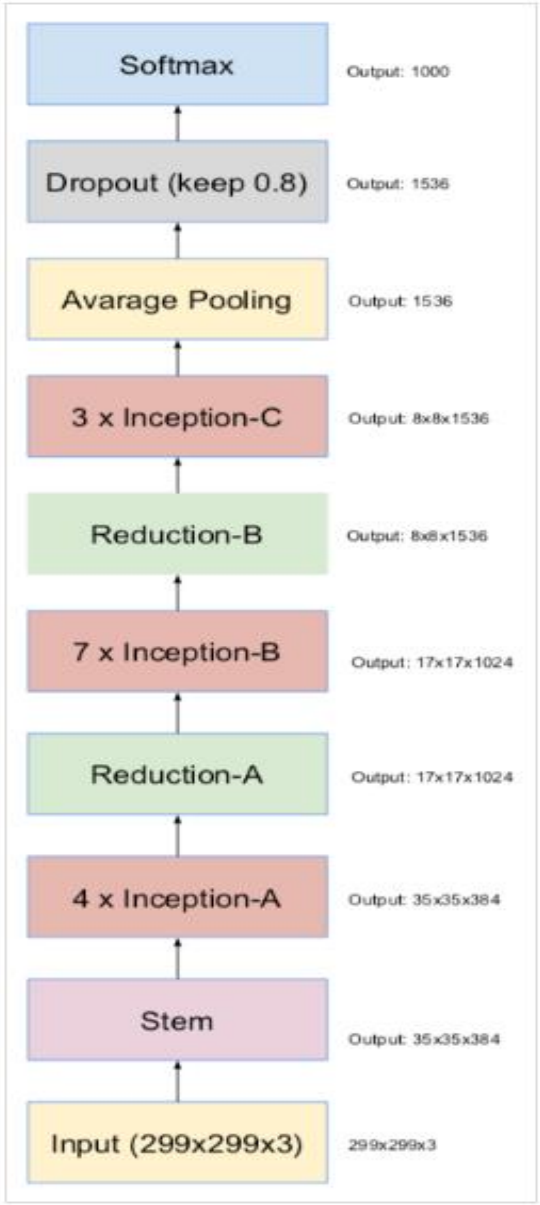
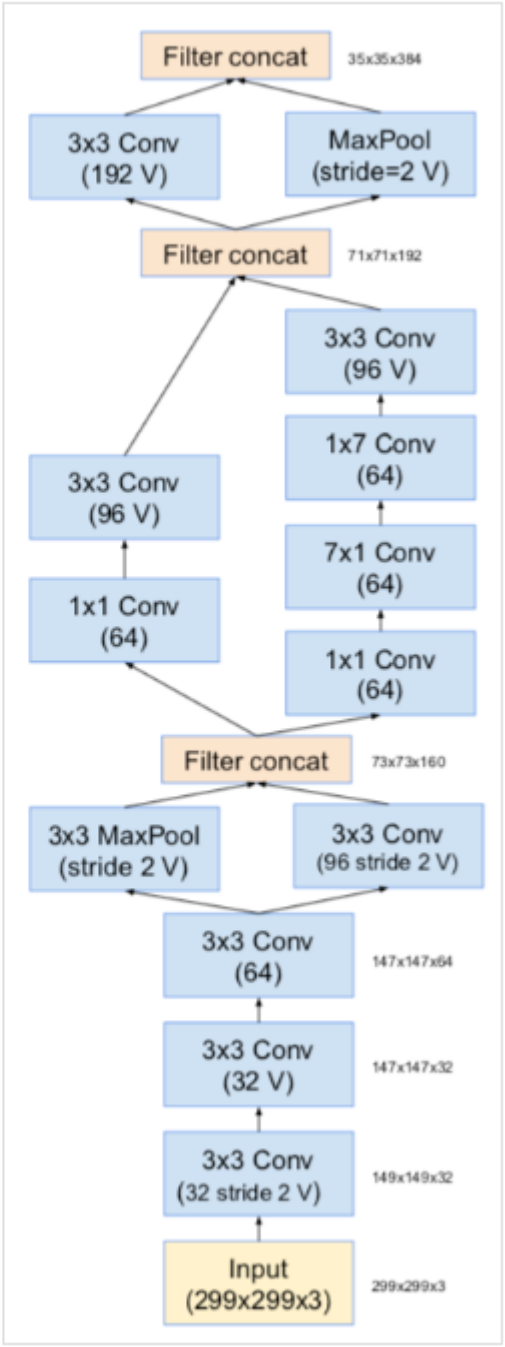
Inception



Stem

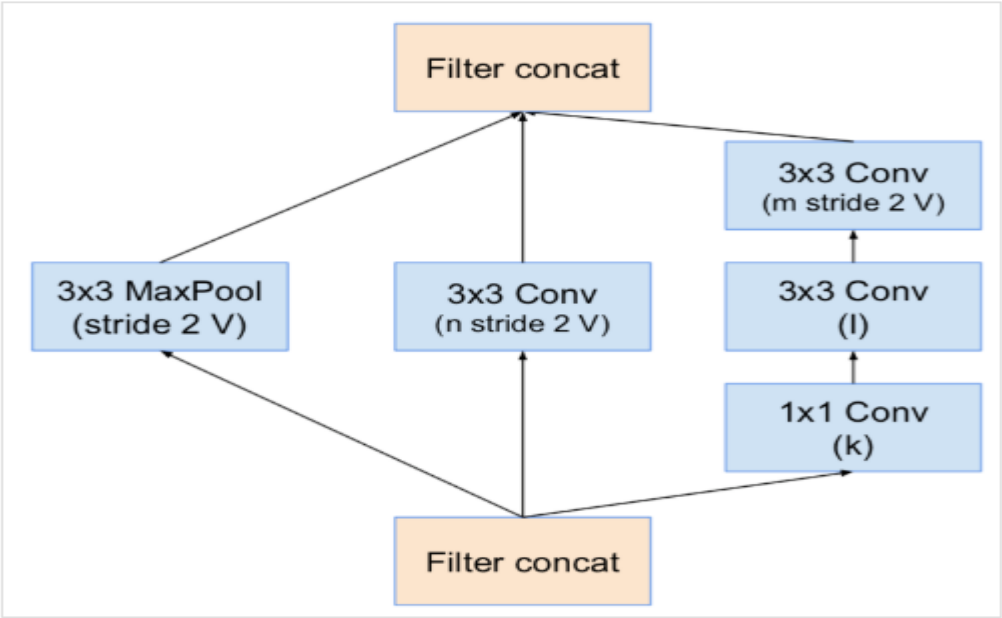
Stem (299x299x3 → 35x35x384)



Inception

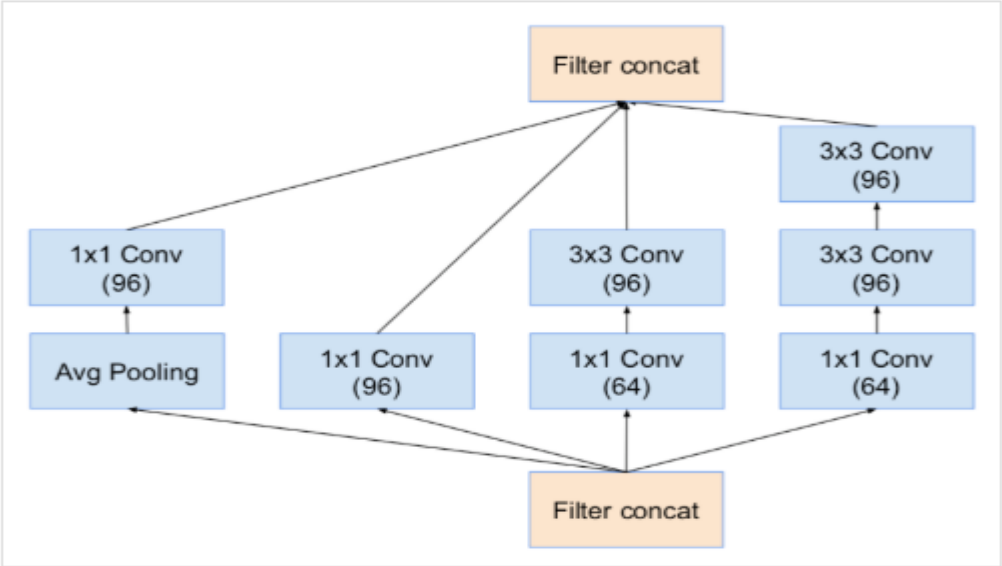
Reduction-A

Reduction-A (35x35x384 → 17x17x1024)

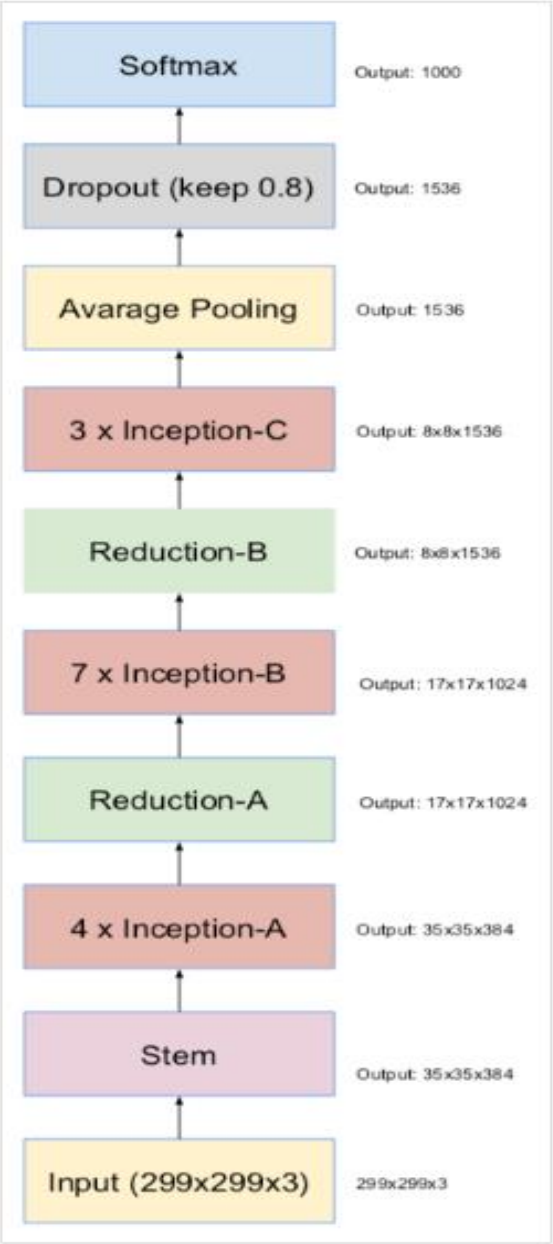


Inception-A

Inception-A (35x35x384 → 35x35x384)

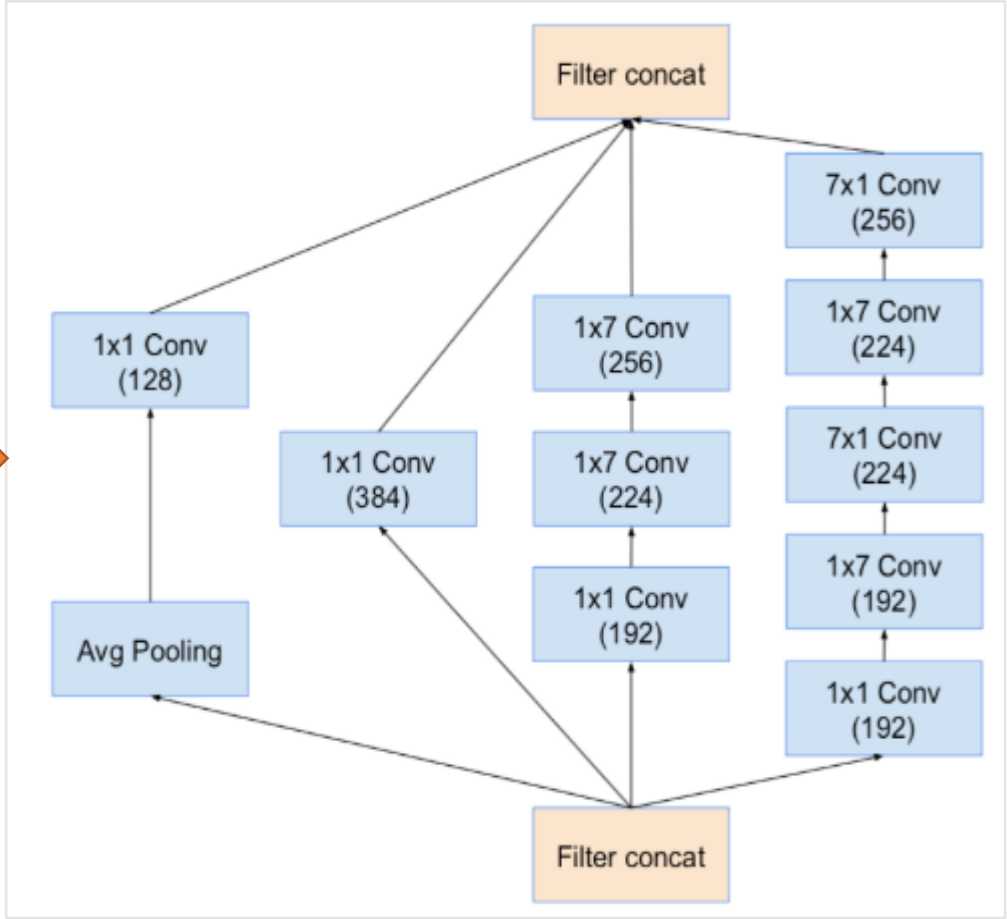


Inception



Inception-B

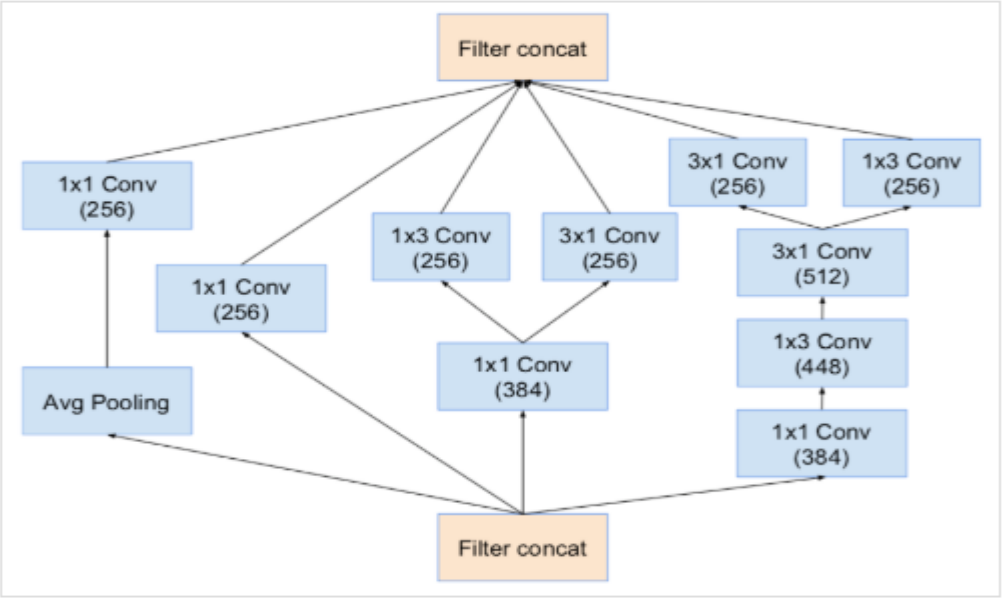
Inception-B (17x17x1024 → 17x17x1024)



Inception

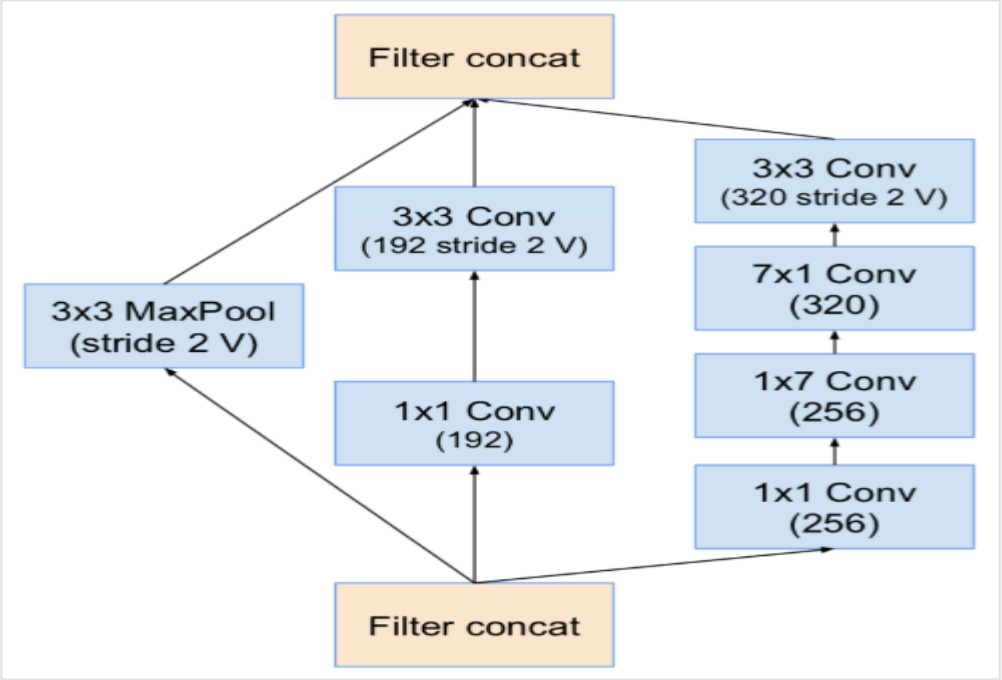
Inception-C

Inception-C (8x8x1536 → 8x8x1536)



Reduction-B

Reduction-B (17x17x1024 → 8x8x1536)



Deeplab-v3-ASPP

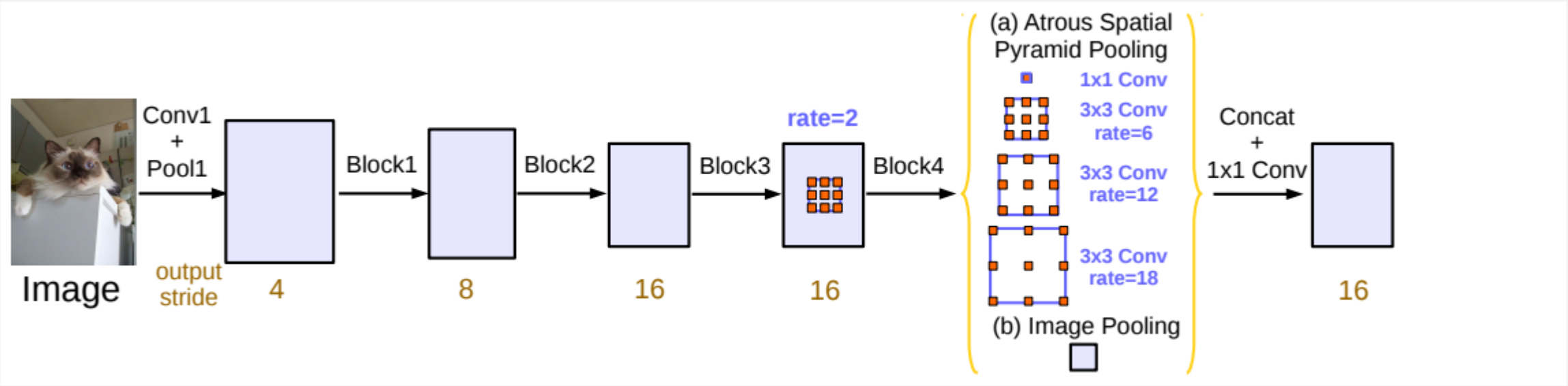
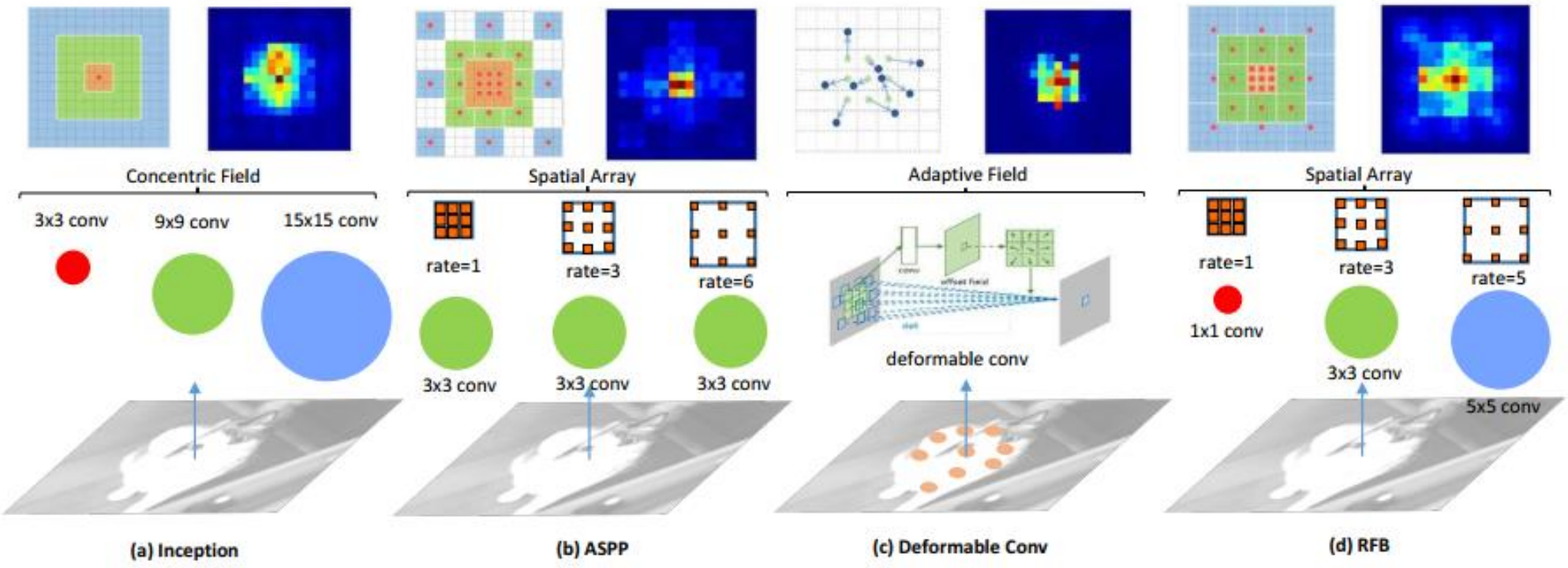


Figure 5. Parallel modules with atrous convolution (ASPP), augmented with image-level features.

Deformable-CNN

Receptive Field Block(RFB)



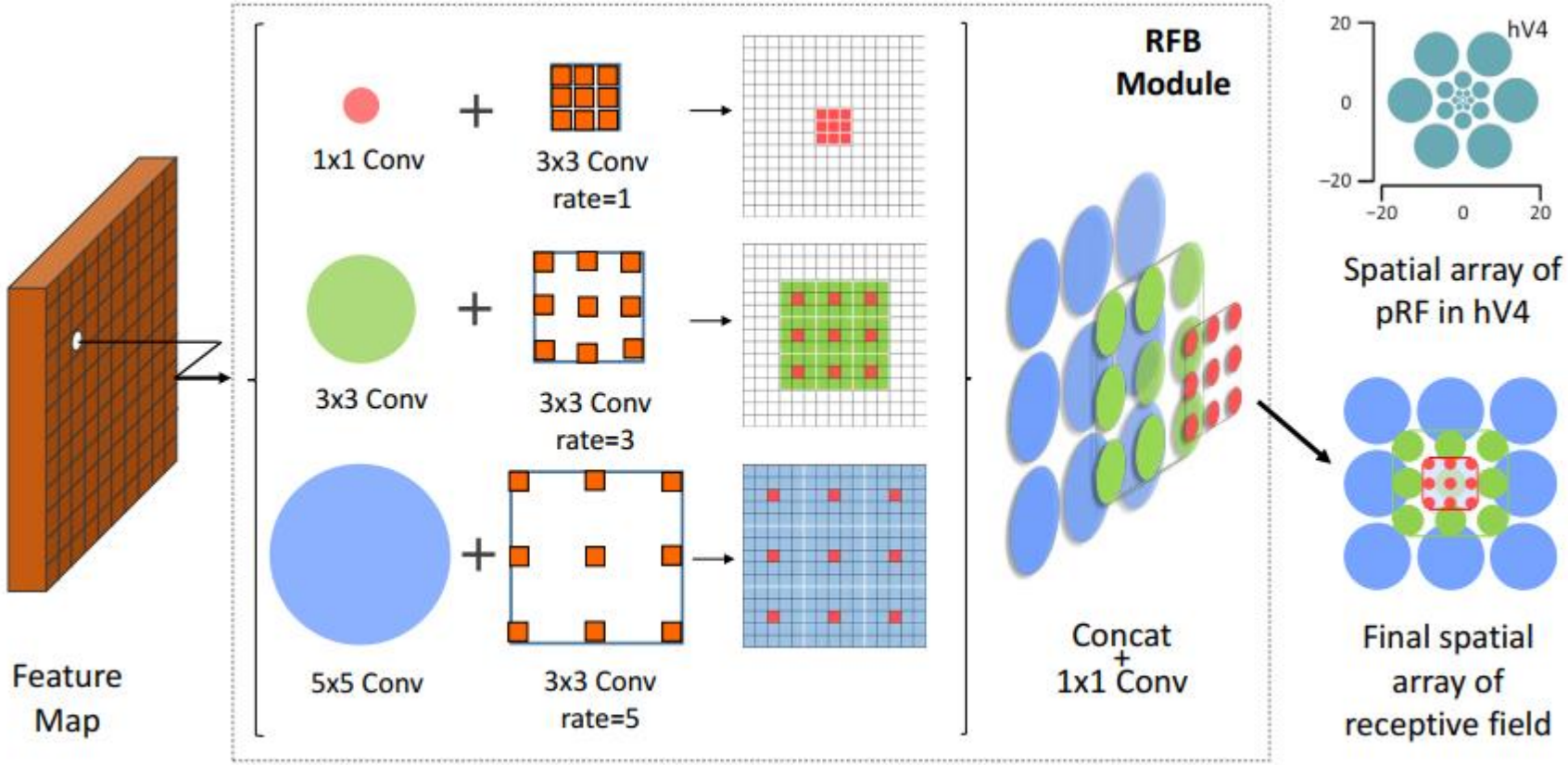
Drawbacks

1. All kernels in Inception are sampled at the same center. --inception

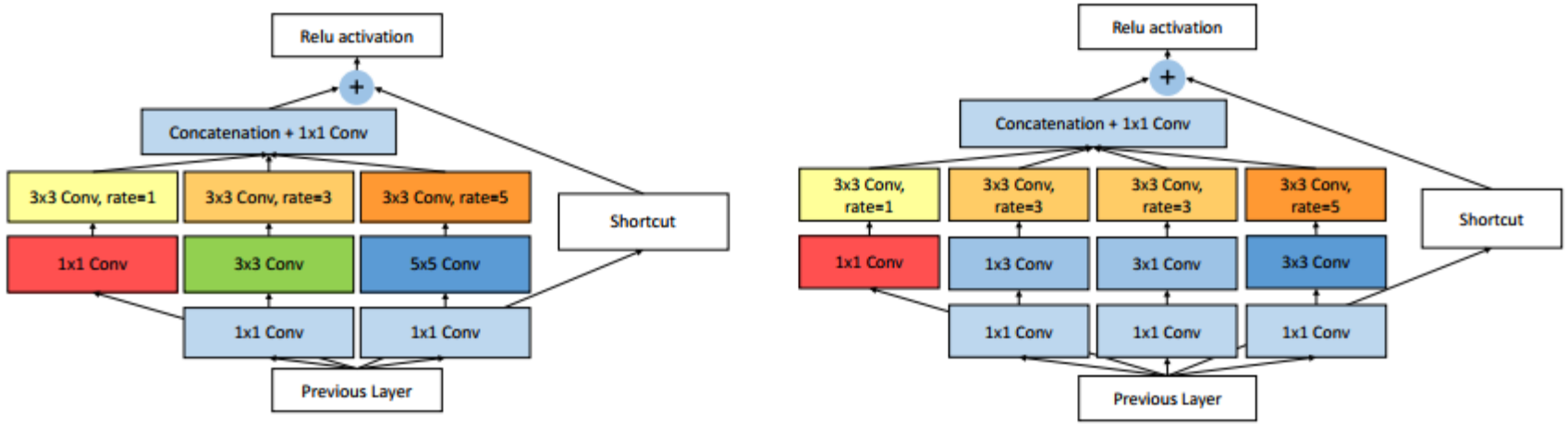
2. But the features only have a uniform resolution from previous convolution layers of the same kernel size, and compared to the daisy shaped ones, the resulting feature tends to be less distinctive. --ASPP

3. Although its sampling grid is flexible, the impact of eccentricity of RFs is not taken into account, where all pixels in an RF contribute equally to the output response and the most important information is not emphasized. --Defor-convnet

Receptive Field Block(RFB)



Receptive Field Block(RFB)



(a) RFB (b) RFB-s

RFB-net

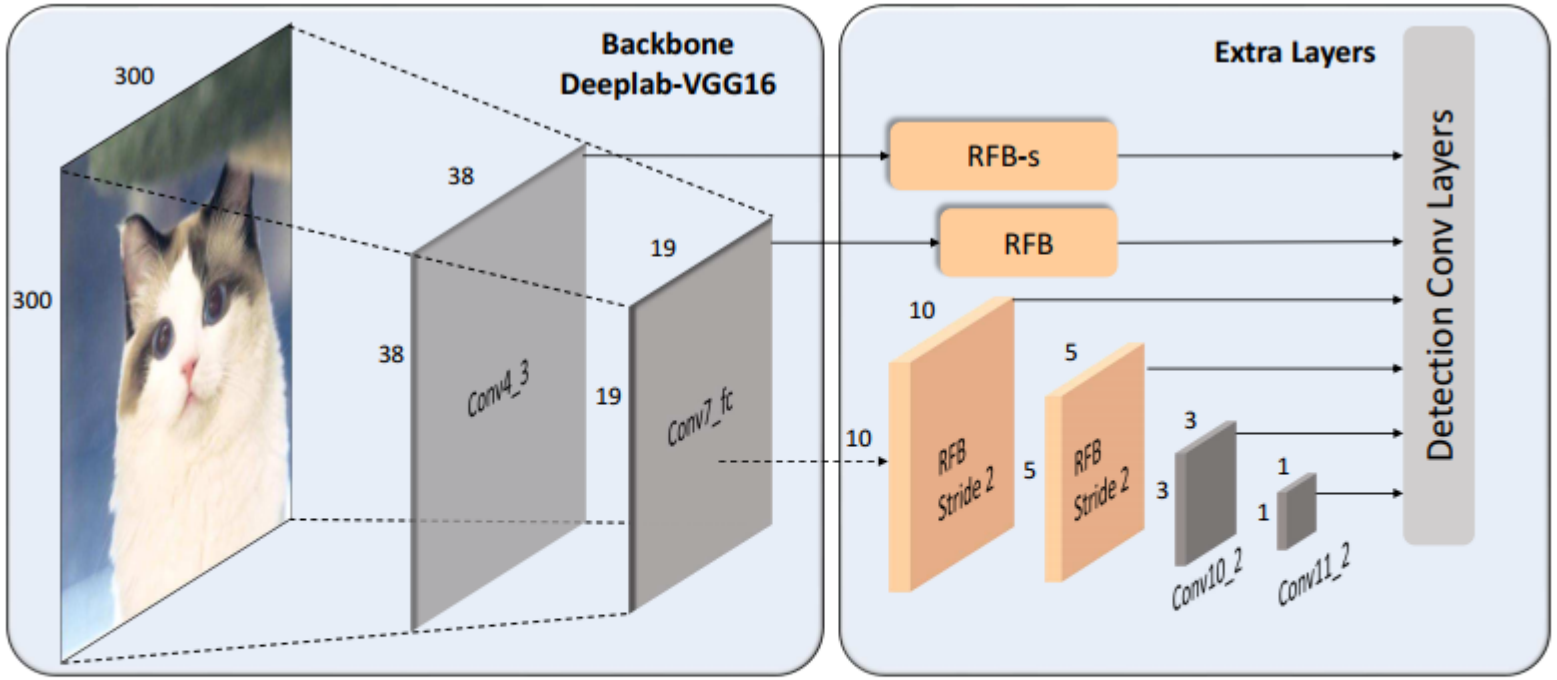


Fig. 5. The pipeline of RFB-Net300. The conv4_3 feature map is tailed by RFB-s which has smaller RFs and an RFB module with stride 2 is produced by operating 2-stride multi-kernel conv-layers in the original RFB.

Experiments

Method	Backbone	Data	mAP(%)	FPS
Faster [26]	VGG	07+12	73.2	7
Faster [11]	ResNet-101	07+12	76.4	5
R-FCN [17]	ResNet-101	07+12	80.5	9
YOLOv2 544 [25]	Darknet	07+12	78.6	40
R-FCN w Deformable CNN [4]	ResNet-101	07+12	**82.6**	8†
SSD300* [22]	VGG	07+12	77.2	**120**‡
DSSD321 [6]	ResNet-101	07+12	78.6	9.5
RFB Net300	VGG	07+12	**80.5**	**83**
SSD512* [22]	VGG	07+12	79.8	50‡
DSSD513 [6]	ResNet-101	07+12	81.5	5.5
RFB Net512	VGG	07+12	**82.2**	**38**

† Extrapolated time

‡ Tested in Pytorch-0.3.0 and CUDNN V6 for fair comparison

Table 1. Comparison of detection methods on the PASCAL VOC 2007 *test* set. All runtime information is computed on a Graphics card of Geforce GTX Titan X (Maxwell architecture).

Experiments

Method	Backbone	Time	Avg. Precision, IoU: 0.5:0.95	0.5	0.75	Avg. Precision, Area: S	M	L
Faster [26]	VGG	147 ms	24.2	45.3	23.5	7.7	26.4	37.1
Faster+++ [11]	ResNet-101	3.36 s	34.9	55.7	37.4	15.6	38.7	50.9
Faster w FPN [19]	ResNet-101-FPN	240 ms	36.2	59.1	39.0	18.2	39.0	48.2
Faster by G-RMI [15]	Inception-Resnet-v2 [31]	–	34.7	55.5	36.7	13.5	38.1	52.0
R-FCN [17]	ResNet-101	110 ms	29.9	51.9	–	10.8	32.8	45.0
R-FCN w Deformable CNN [4]	ResNet-101	125ms†	34.5	55.0	–	14.0	37.7	50.3
Mask R-CNN [9]	ResNext-101-FPN	210 ms	37.1	60.0	39.4	16.9	39.9	53.5
YOLOv2 [25]	darknet	25 ms	21.6	44.0	19.2	5.0	22.4	35.5
SSD300* [22]	VGG	12 ms‡	25.1	43.1	25.8	–	–	–
SSD512* [22]	VGG	28 ms‡	28.8	48.5	30.3	–	–	–
DSSD513 [6]	ResNet-101	182 ms	33.2	53.3	35.2	13.0	35.4	51.1
RetinaNet500 [20]	ResNet-101-FPN	90 ms	34.4	53.1	36.8	14.7	38.5	49.1
RetinaNet800 [20]	ResNet-101-FPN	198 ms	**39.1**	59.1	42.3	21.8	42.7	50.2
RFB Net300	VGG	**15 ms**	**30.3**	49.3	31.8	11.8	31.9	45.9
RFB Net512	VGG	**30 ms**	**33.8**	54.2	35.9	16.2	37.1	47.4
RFB Net512-E	VGG	**33 ms**	**34.4**	55.7	36.4	17.6	37.0	47.6

† Extrapolated time
‡ Tested in Pytorch-0.3.0 and CUDNN V6 for fair comparison

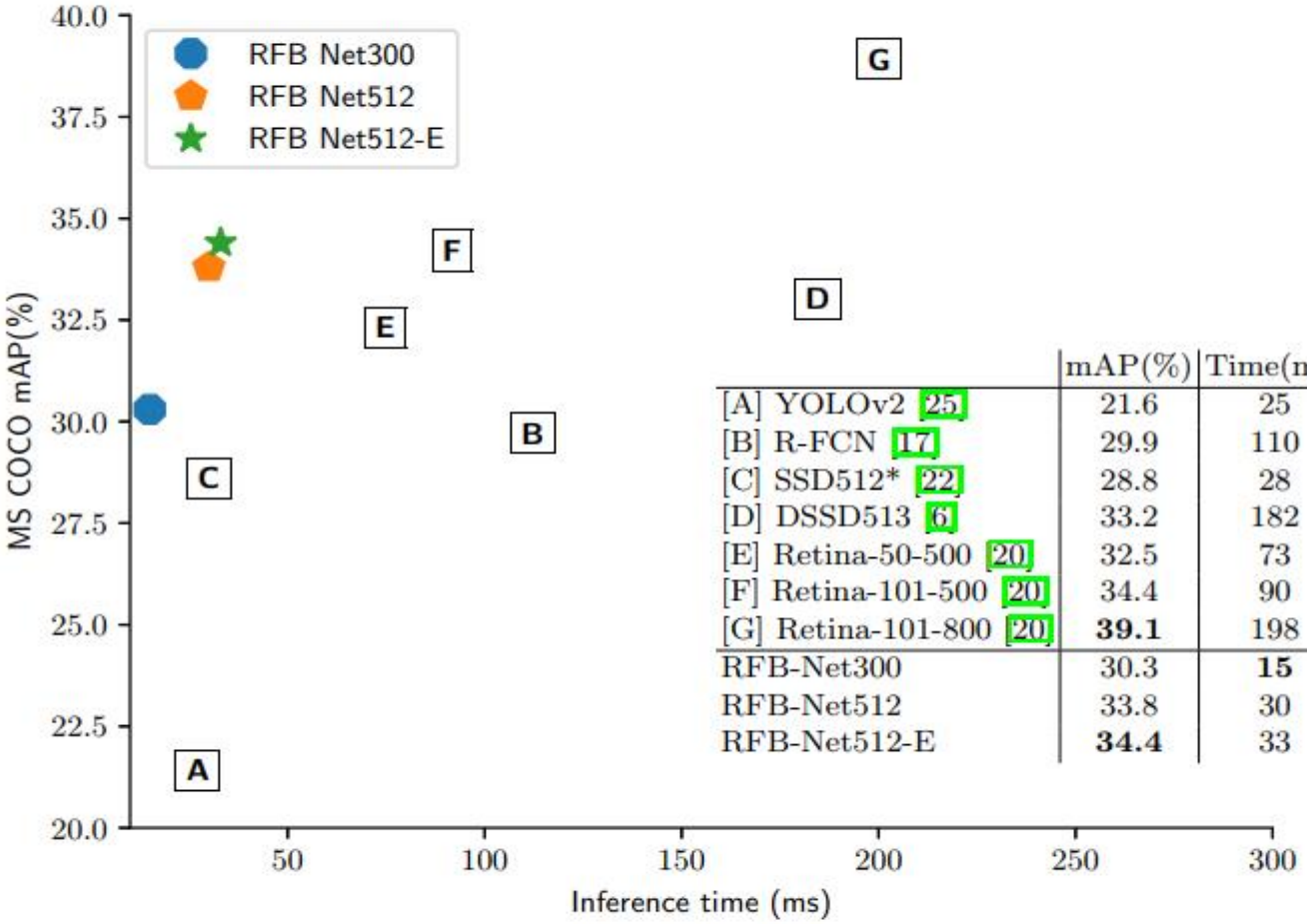
Table 4. Detection performance on the COCO *test-dev* 2015 dataset. Almost all the methods are measured on the Nvidia Titan X (Maxwell architecture) GPU, except RetinaNet, Mask R-CNN and FPN (Nvidia M40 GPU).

Experiments

Architecture	#parameters	VOC 2007 mAP (%)	COCO minival mAP (%)
RFB	34.5M	**80.1**	**29.7**
Inception [33]	32.9M	78.4	27.3
Inception-L	33.3M	79.5	28.5
ASPP-S	33.4M	79.7	28.1
Deformable CNN [4]	35.2M	79.5	27.6

Table 3. Comparison of different blocks on VOC 2007 *test* and MS COCO *minival2014*.

Experiments



	mAP(%)	Time(ms)
[A] YOLOv2 [25]	21.6	25
[B] R-FCN [17]	29.9	110
[C] SSD512* [22]	28.8	28
[D] DSSD513 [6]	33.2	182
[E] Retina-50-500 [20]	32.5	73
[F] Retina-101-500 [20]	34.4	90
[G] Retina-101-800 [20]	**39.1**	198
RFB-Net300	30.3	**15**
RFB-Net512	33.8	30
RFB-Net512-E	**34.4**	33

Fig. 6. Speed (ms) vs. accuracy (mAP) on MS COCO *test-dev*. Enabled by the proposed RFB module, our single one-stage detector surpasses all existing high frame rate detectors, including the best reported one-stage system Retina-50-500 [20].

Conclusions

1. New backbone to find feature?
2. How?
3. Find what?